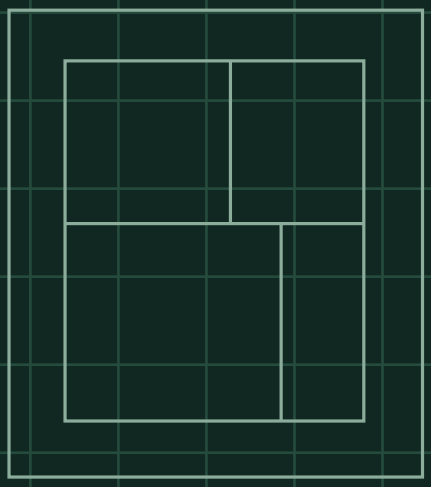
AMMARAWATI / CONSTRUCTION INTELLIGENCE

PLATFORM ARCHITECTURE & BUILD GUIDE

E-commerce / Customer intelligence / Land planning
Material pricing / Engineering knowledge / Replaceable AI



A DEVELOPER-READY PRODUCT BLUEPRINT

- Public storefront + two admin portals
- Private PostgreSQL + company-owned cloud
- Nepal-first planning with professional review
- Build scope, UI copy, data model, APIs and tests

VERSION 1.0 | 26 OCTOBER 2026

Planning document - not approved architectural or structural drawings

Reader’s map

A single working specification for the product owner, UI designer, frontend team, backend team, database engineer, security reviewer and construction-domain reviewers. Decisions not yet made are flagged instead of guessed.

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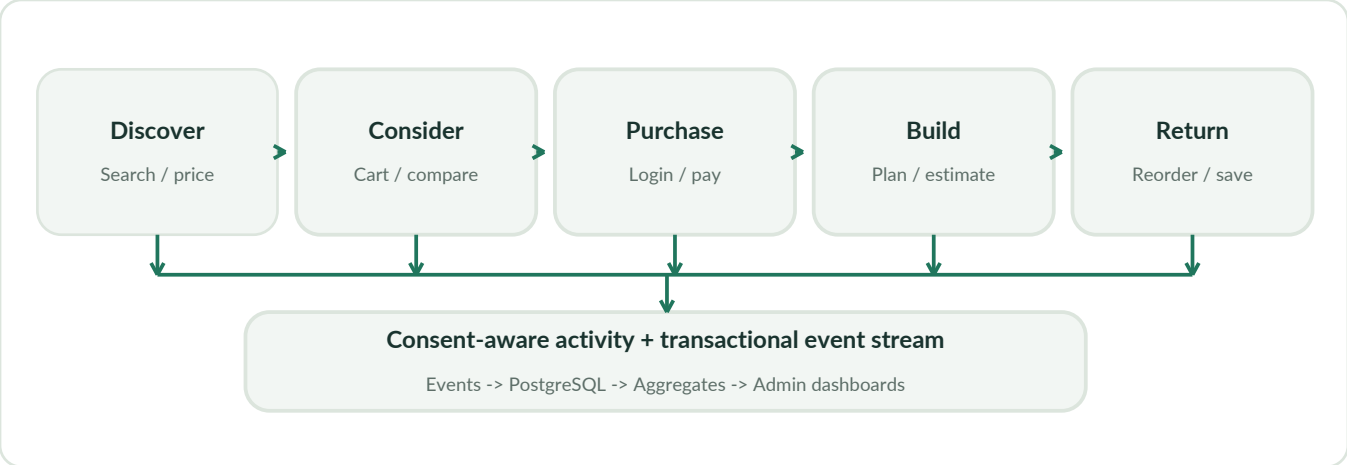
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HOW TO USE THIS GUIDE

Start with the phase-one scope, then use the site map and UI specification to design screens. Build identity, commerce, events, prices and regional permissions against the data/API contract. Treat Nepal building-rule data and material rates as operator-approved content, not generated facts.

01 / Executive product brief

Vision. A product-first building-supply marketplace similar in shopping journey to The Builders Supply, combined with land-to-building concepts, material education, preliminary estimating, and a permission-aware customer intelligence platform. The reference site demonstrates category/brand browsing, product cards, prices, inventory signals and purchase actions [R1].



User promise	System delivers
Shop without friction	Guests can browse catalog, search SKU/brand, see published prices, compare products and use a cart.
Buy with an account	Login to complete secure checkout, receive order records and track delivery.
Plan what to build	Logged-in customers enter land details, see unapproved 2D concepts and learn preliminary cost components.
Run the business	Super Admin has company-wide controls; Regional Admin handles only assigned territories.
Own the intelligence	All authoritative customer, catalog, order, rate and project records live in company-controlled private PostgreSQL.

NON-NEGOTIABLES

No automated claim of municipal approval or structural safety; no invented prices; no unrestricted administrator access; no non-essential behavioral profiling before the required notice/choice; and no LLM authority to change production data.

02 / Scope, launch priority and missing inputs

Now: launch candidate	After launch	Deliberately later
Public catalog, product detail, search and filters	Advanced search/recommendations	Full BIM/CAD and permitted drawings
Guest cart; login-based checkout; orders	Supplier/warehouse integrations	Company-trained proprietary LLM
Secure Super and Regional portals	Advanced geographic analytics	Active-active multi-cloud databases
PostgreSQL tracking and consent controls	Irregular plots, advanced land CAD	Automated legally certified engineering
Initial land planner, 2D concepts, material guides	Expanded municipality rule coverage	Complex commercial engineering automation
Operator rates, conceptual estimates, AI explanations	AI-assisted BOQ after verified drawings	Unreviewed autonomous cost or design changes

Unresolved decisions before production launch

- **Commercial:** supplier ownership model, product catalog, fulfillment territories, tax, shipping, returns and payment provider.
- **Location:** initial Nepal municipalities, applicability of current bylaws, approved professional reviewer(s).
- **Pricing:** rate owners, currency, authoritative regional sources, rate approval thresholds, cost accuracy bands.
- **Operations:** customer support contacts, consent/retention policy, cloud account and incident responders.
- **Brand:** official domain, logo, photography permission, Nepali/English copy approval.

03 / Information architecture and full site map

Public storefront: navigation and routes

Navigation group	Route patterns and pages
Discover & shop	/ /shop /categories /category/[slug] /brands /search?q= /product/[slug] /compare
Cart & checkout	/cart /login /register /checkout /checkout/confirmation /orders/[id]
Construction knowledge	/construction /materials /materials/[slug] /guides /guides/[slug] /standards/nepal
Land / building planner	/plan /plan/land /plan/requirements /plan/concepts /plan/concept/[id] /plan/estimate
Cost & AI	/cost-estimator /cost-methodology /assistant /consultation
Customer account	/account /account/orders /account/projects /account/estimates /account/privacy
Company and legal	/about /contact /faq /delivery /returns /privacy /terms /engineering-disclaimer

Separate admin website maps

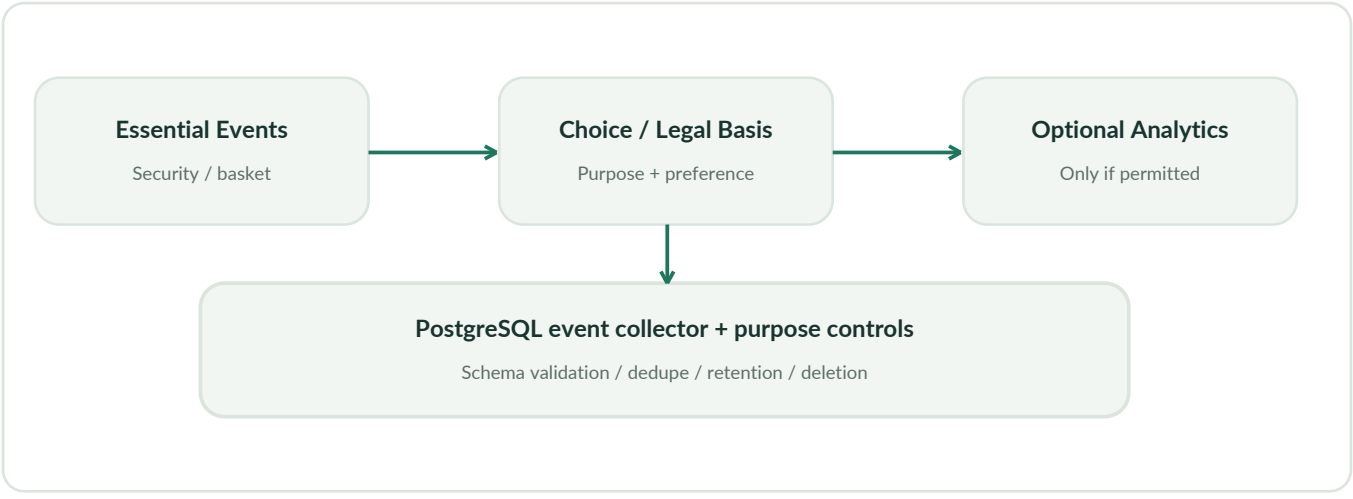
Portal	Primary navigation
Super Admin	Overview / Catalog / Categories & brands / Price master / Rate approvals / Orders / Inventory / Customers / Segments / Construction content / Plan templates / Municipality rules / AI providers / Regional teams / Audit & security / Settings
Regional / Operations Admin	My region / Enquiries / Assigned customers / Orders / Fulfillment / Local inventory / Rate proposals / Plan reviews / Quotes / Demand reports / Team tasks

Site rule: Three independently deployed Next.js apps, separate sign-in surfaces, shared visual design tokens, one authenticated backend. Admin URLs are protected by stronger access controls; no admin UI directly connects to PostgreSQL.

04 / Customer journeys and access model

Moment	Guest visitor	Logged-in customer
Search / product	Search, filter, browse and see public prices	Same plus personalized capabilities where permitted
Cart	Add/update/remove lines using essential cart token	Cart linked to account after secure merge
Checkout	Prompt to sign in/create account	Validated address, payment and confirmation
Construction guides	Access general educational guidance	Access contextual and saved learning
Land planning	Preview tool and non-stored demo	Create projects, drawings, preliminary estimates
Data controls	Analytics preferences and notice	Review privacy preferences, export/delete requests

Core activity/event sequence



Purchase completion is server-authoritative. A browser-side event is never proof of payment; the backend records order status after validating payment notifications, checking signatures and applying idempotency controls.

05 / UI/UX style system

Token	Proposed specification
Visual direction	Architectural, clean, credible. Large whitespace, warm neutrals, purposeful drawings and real material photography.
Palette	Evergreen #20765D; dark ink #19332E; ivory #FBF7F0; pale sage #F2F6F3; copper accent #C78945.
Fonts	Inter/Lato for English; tested Devanagari font for Nepali; consistent type scale.
Grid	12 columns desktop, 8 tablet, 4 mobile. Maximum content width 1,280 px.
Spacing	8 px base scale; 44+ px minimum target sizing where applicable; consistent responsive density.
Product cards	Photo, name, SKU, specification hints, public price, availability, compare and add-to-cart.
Planning UI	Step-by-step wizard, progress, editable assumptions, readable SVG line drawings and warnings.
Quality	WCAG 2.2 AA target; tested contrast; keyboard navigation; screen-reader labels; low-bandwidth behavior.

Public homepage: annotated text wireframe

Area	Visible content / action
UTILITY BAR	Delivery locations / Contact / Help / Language switch
HEADER	AMMARAWATI logo | Search products, brands or SKU | Account | Cart
CATEGORY NAV	Structural Exterior Interior Electrical Plumbing Tools Building Planner
HERO	Find the right materials for every project. [Shop Materials] [Plan My Building]
DISCOVERY	Shop by Category | Popular Products | Recommended Materials
TRUST	Specifications / Clear pricing / Order support / Local availability
PLANNING	Have land? Explore 2D concepts, materials and preliminary construction costs.
CONTENT	How-to guides / Nepal building information / Material comparison
FOOTER	Delivery, returns, privacy, construction disclaimer, contact

Product page and mobile behavior

- **Desktop product page:** gallery left; title, SKU, visible approved price, unit, stock/location, quantity and cart action right; specifications and related items below.
- **Mobile:** simplified logo/search/cart header; category drawer; full-width product cards; sticky product purchase bar; single-column planning wizard.
- **Empty / error / loading states:** clear no-results suggestions, unavailable-rate messages, recoverable cart errors and save progress.
- **Design deliverables:** Figma tokens, desktop/tablet/mobile component variants, keyboard interactions, responsive states, content standards.

06 / Approved starting public copy

BRAND LINE

AMMARAWATI CONSTRUCTION - Everything for Construction. From Materials to Planning.

Page element	Proposed copy
Hero heading	Find the Right Materials. Build with Confidence.
Hero support	Browse building supplies, compare published prices, and explore the information you need before starting your next project.
Primary CTA	Shop Building Materials
Secondary CTA	Plan My Building
Search placeholder	Search products, brands, materials or SKU...
Building planner	Have land? Explore early building possibilities, material options and a preliminary project budget.
Material library	Understand what a material is used for, what specifications matter, and which alternatives may suit your project.
Cost engine	Explore a breakdown based on approved available rates and stated assumptions; final quotes require verification.
Login benefit	Save projects, manage purchases, compare materials and return to your estimates anytime.
Assistant	Ask about building materials and construction steps. We explain sources and flag decisions requiring an engineer.
Footer notice	Concept plans and indicative costs are for early planning only; they are not certified design documents or binding quotations.

Ready-to-use homepage section paragraphs

Shop by Category. Find construction essentials for the work ahead. Browse structural supplies, exterior products, interior finishes, tools and building services. Compare specifications and available prices before adding products to your cart.

Building Planner. Already have land? Begin with your measurements, selected municipality and building goals. Explore early residential or commercial layout ideas, then see what further information an architect or engineer will need.

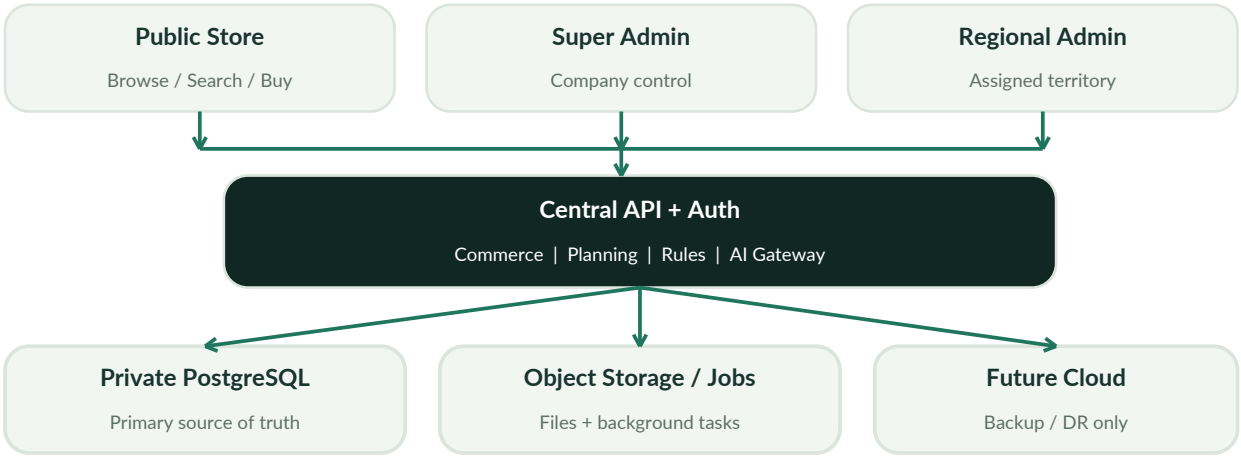
Understand Your Budget. View the major cost elements of your project, from foundations to finishes. Estimates use approved available regional rates and show the assumptions and exclusions behind each total.

Trust and limitations. Prices, stock availability and local regulations can change. Construction concepts are educational and preliminary; certified designs and building permits require professional and authority review.

Practical page copy and SEO guidance

- **Category intro:** “Browse [category] materials by specification, application, availability and approved price.”
- **Price not ready:** “Current price unavailable for your selected area. Request an updated quote.”
- **Checkout:** “Review delivery, taxes and payment before placing your order.”
- **Plan review:** “This sketch is illustrative. Local bylaws and professional design approval must be checked.”
- **SEO title:** “Ammarawati Construction | Materials, Building Planning & Cost Estimates.”
- **SEO description:** “Explore construction materials, compare published prices, shop supplies, and plan residential or commercial projects.”

07 / Technical stack and system topology



Independent frontends; API permissions enforced server-side. Additional cloud is optional.

Layer	Baseline / rationale
3 browser apps	Next.js + TypeScript for shop, Super Admin and Regional Admin.
API	NestJS modular monolith with domain boundaries, validation and versioned contracts.
Primary database	Privately deployed PostgreSQL; PostGIS when geospatial queries are needed.
Async work	PostgreSQL outbox + workers; Redis/job queue only if operationally warranted.
Files	Company-controlled encrypted object storage for catalogs, photos, PDFs and generated concepts.
Search	PostgreSQL FTS at launch; optional dedicated search index if performance/scale warrants.
Auth	OIDC-compatible identity service, phishing-resistant administrator MFA, scoped roles.
Observability	OpenTelemetry metrics/traces, centralized audit logs, security alerts.
Cloud independence	Container images + infrastructure-as-code + restore-tested backup portability.
LLM	One AI Gateway with replaceable external or self-hosted model adapter.

Monorepo build structure

apps/ public-web, super-admin, regional-admin; **services/** api, workers; **packages/** ui, design-tokens, api-contracts, validation, events; **database/** migrations, seeds, tests; **infra/** terraform-or-opentofu, containers, monitoring; **tests/** unit, integration, permission, e2e, security.

08 / Backend domains and interface contracts

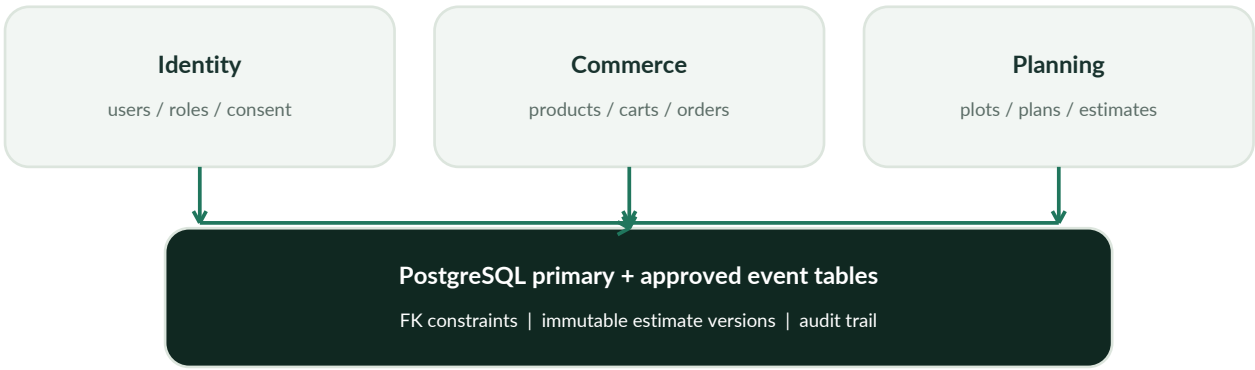
Domain	Key responsibilities
Identity & organizations	Customers, teams, login, roles, region scopes, MFA, consent and sessions.
Catalog & merchandising	Categories, brands, SKUs, variants, attributes, product media, content.
Pricing & rates	Retail prices, regional material rates, approval, validity and historical versions.
Commerce	Carts, checkout, orders, payments, returns, shipping and stock reservations.
Planning	Plot profiles, dimensional geometry, design options, plan validity and savings.
Costs	Work item rates, quantity methods, estimate snapshots, assumptions and reports.
Knowledge / AI	Technical sources, review state, restricted tool retrieval, explanations.
Telemetry / analytics	Consented events, aggregation, cohorts and region reports.

Minimum API contract: example routes

Route	Expected behavior
GET /api/v1/products?query=&region;=	Public indexed catalog, approved visible rates and pagination
GET /api/v1/products/:id	Product specs, images, rate availability, stock policy
POST /api/v1/cart/items	Validated, rate-limited guest/account cart mutation
POST /api/v1/checkout	Reprice & reserve server-side, create pending order
POST /api/v1/payments/webhook	Verify signature; replay-safe order payment update
GET /api/v1/me/orders	Customer-owned orders only
POST /api/v1/plans	Save validated plot data and building brief
POST /api/v1/plans/:id/concepts	Return unapproved deterministic SVG geometry options
POST /api/v1/estimates	Price versioned quantities from approved local rates
POST /api/v1/events	Accept schema-conforming events only for allowed purposes
POST /api/v1/admin/rates/:id/approve	Role- and region-authorized rate approval
GET /api/v1/admin/analytics/regions	Aggregated, scoped regional funnel reports

API contract rules: requests use typed schemas; IDs are opaque; objects are owner/region checked; all write operations have idempotency where required; errors do not expose secrets or private information. Generate OpenAPI specifications and an API client for all three frontend apps.

09 / PostgreSQL data model



Schema / tables	Minimum essential fields and constraints
identity.users, roles, region_assignments	UUID IDs, identity reference, state, MFA markers, region scopes; unique constraints.
privacy.consent_records	subject/session, purpose, notice_version, choice, recorded_at, source, withdrawal_at.
catalog.products, variants, prices	SKU unique, attributes, images, unit, currency, region, price_version and active dates.
commerce.carts, cart_items	session/account owner, item, quantity, validated session state.
commerce.orders, order_items, payments	state machine, signed payment reference, item price snapshots, immutable totals.
commerce.warehouses, stock_movements	location, stock reservation, adjustment actor, concurrency guards.
projects.plots, requirements, concepts	polygon/dimensions, address area, road access, option SVG, rule_version, validity.
costs.work_items, material_rates	measurement method, cost_unit, currency, geo scope, rate state, approved_by.
costs.estimates, estimate_lines	amount decimals/minor units, quantity, rate snapshot, exclusions, assumptions.
knowledge.rules, documents	jurisdiction, edition, citation, scope, verifier, review_status, valid_dates.
analytics.activity_events, sessions	event_id, purpose, minimal metadata, visitor pseudonym, retention deadline.
security.audit_events, policy_versions	actor, action, target, timestamp, affected scope, immutable append process.

Data design and ownership

- **Primary source of truth:** one PostgreSQL writer. Customer, product, transaction, rate and project facts remain authoritative here.
- **Isolation:** restrict apps through APIs; PostgreSQL row-level policies where appropriate; encrypted storage, no public DB listener.
- **History:** published material rates and finalized estimates are versioned; historical reports cannot silently reprice.
- **Data quality:** foreign keys, unique keys, audit metadata, required source fields and migrations in source control.
- **Future:** add optional read models or other databases through approved event interfaces; never introduce two simultaneous uncontrolled order writers.

10 / Customer event tracking and privacy

The site can establish technically necessary session, cart and security operations at arrival. **Optional browsing analytics, personalization and profile-building must be gated by the privacy rules that apply to the actual market and users.** No silent cross-device identity stitching. Nepal-specific privacy and e-commerce obligations and other applicable regimes require local legal assessment [R6].

Event	Essential example fields	Meaning
search_submitted	event_id, query_category, result_count, filter_ids	Product demand; not evidence of purchase
product_viewed	product_id, listing_context	Product discovery
price_viewed	product_id, published_price_version	Price interest, not price acceptance
cart_item_added	cart_id, product_id, quantity	Potential purchasing intent
checkout_started	cart_id, checkout_id	Funnel stage
order_paid (server)	order_id, paid_amount, currency	Validated transaction only
plan_created	project_type, region_code, area_band	Declared construction project
material_compared	material_ids, compare_context	Interest in alternatives
estimate_saved	estimate_id, level, budget_band	Planning intent

Event acceptance and privacy safeguards

- Collect only allowlisted fields; exclude names, detailed addresses, payment secrets and unfiltered chat text from analytics.
- Store consent notice version, purpose and timestamp; enforce choices in both browser collector and API.
- Separate **declared preference**, **observed interest** and **confirmed transaction**.
- Identify visitors via limited pseudonymous session identifiers, with short retention as justified and regional aggregation.
- Grant analysts aggregated reports by default; audit individual-record access and suppress tiny identifiable groups.
- Provide required data export, correction, withdrawal and deletion workflows without deleting legally required accounting records.

Dashboards and business metrics

Funnels: search-to-product, product-to-price, price-to-cart, cart-to-checkout, checkout-to-paid; by date, catalog, device and selected service region. Operational KPIs: gross sales, order count, refund rate, out-of-stock search demand, no-result terms and regional building-project inquiries. Report both unique users/sessions and event counts.

11 / Commerce operations and fulfillment

Product content and pricing

Each SKU/variant must have category, brand, unit, specification, images, availability, published sell price, currency, tax class, applicable geography and operational status. Admins may maintain multiple rate lists, but only approved active public prices are exposed. Consumer retail price and construction-estimating material cost may differ; label them distinctly.

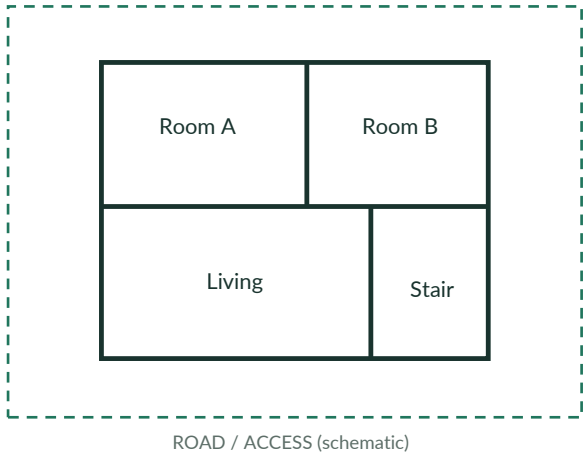
Checkout state machine

Cart -> Draft Checkout -> Pending Payment -> Paid -> Fulfillment -> Delivered / Returned / Refunded. The server recalculates item rates, delivery and tax; reservations are transaction-safe; payment callbacks are verified and replay-safe. Do not store raw bank-card credentials. Apply the chosen payment provider's security and compliance requirements.

Operational screens

Screen	Must support
Catalog editor	SKU variants, specifications, media, stock visibility, rate source and publish status.
Order queue	Payment state, address, stock reservations, picking, shipment, exceptions and refund permissions.
Regional fulfillment	Filter by authorized warehouses/territory, validate transitions, show assigned staff.
Returns	Reason codes, condition assessment, customer notices, refund reconciliation and audit trail.
Inventory	Movement ledger, pending reservations, stock counts, safe concurrent adjustments.

12 / Nepal land-to-building concept planner



INPUT: polygon, road side,
municipality, floors, use

RULE CHECK: verified bylaw
or mark unverified

OUTPUT: room schedule,
SVG line sketch, assumptions

NOT: an approved naksa pass

The first planner accepts **plot area plus measured frontage/depth or polygon points**, orientation, adjacent roads, municipality/ward, plot use, likely number of floors and residential/commercial requirements. Nepal-friendly land units include ropani-aana-paisa-daam and bigha-kattha-dhur, alongside ft² and m². Convert units deterministically and retain the original entry.

Planning algorithm: concept-only

Step	Validation / output
1. Validate plot	Area, shape, boundary dimensions, road side, orientation and missing measurements.
2. Resolve jurisdiction	Match local zoning/bylaw edition; indicate missing or unverified legal data.
3. Build envelope	Apply verified setbacks/coverage/height constraints; do not guess missing values.
4. Assign room zones	Use residential or commercial approved template sets and circulation assumptions.
5. Generate 2D SVG	Dimensioned concept only where inputs and verified constraints permit; show scale, north and labels.
6. Review constraints	Flag daylight, ventilation, egress, fire, accessibility and structural-review requirements.
7. Estimate elements	Extract approximate room, wall and roof measures with confidence and exceptions.
8. Save version	Rule version, input snapshot, assumptions, concept SVG and review status.

Nepal regulatory registry: required sources

National guidance includes DUDBC's NBC 206:2024 Architectural Design Requirements and NBC 205:2024 low-rise RCC detailing guidance [R2], while seismic provisions should be checked against the applicable official edition, including the available NBC 105:2025 revision [R3]. Kathmandu's eBPS publishes municipal building-bylaw references and amendments [R4]. **There is no one safe national setback/FAR value.** Store municipality, zoning scope, source URL, effective date, verification actor and review expiry.

Worked example (illustrative only): a customer enters a 30 ft × 40 ft rectangular plot with the road along one 30 ft side and requests a two-floor family residence. The software stores the 1,200 ft² plot area, flags missing setbacks or soil information, proposes room-zoning alternatives only within verified constraints, and creates a materials/cost category list with missing rates clearly marked. No structural sizes or permit eligibility are inferred from those dimensions alone.

ENGINEERING AND PERMIT BOUNDARY

Never market the generated concepts as municipal "naksa pass" approvals or structural drawings. Earthquake design, soil and foundation assessment, fire/egress, building services and permit submissions require qualified professionals and authority approval.

13 / Residential and commercial knowledge library

Topic	Residential path	Commercial path
Program	Bedrooms, kitchens, baths, family/rental floors	Retail, office, warehouse, mixed-use functions
Building services	Domestic electrical, plumbing, water, ventilation	Demand-based HVAC, fire, electrical, water and special systems
Compliance emphasis	Site, structure, light, ventilation, seismic review	Occupancy, fire safety, access, egress, commercial-use rules
Material selection	Exterior envelope, interior finishes, household fixtures	High-use finishes, envelope, partitions, service capacity
Cost management	Construction and finish grade, owner budget	Services, operations, lifecycle and specification allowances

Material detail record format

Name / canonical ID; category; local-language synonyms; manufacturer/grade; unit; technical standard; dimensions; typical use; constraints; compatibility; maintenance; lifecycle considerations; verified source; review date; region; published price availability; related materials. Each engineering assertion must have a review state and technical source.

Example educational modules

- Site preparation, soil investigation and why foundation decisions depend on engineering input.
- Structural systems, reinforced concrete, masonry and steel: applications and decision factors.
- External walls, roofing, waterproofing and climate-responsive choices.
- Internal finishes, partitions, flooring, sanitary and electrical work.
- Building stages, procurement checks, common mistakes and renovation planning.
- Commercial fit-out, building-services coordination, fire/accessibility planning and maintenance.

14 / Estimation engine and operator-controlled rates

The customer can see an indicative construction breakdown before a detailed contractor quotation exists. **Only operators supply, review and approve prices.** An AI model may explain cost differences but must never invent, approve or override rate entries.

Three estimate levels

Level	Inputs / limits	Output
Conceptual	Project type, approximate gross area, region, quality band; major uncertainty.	Elemental ranges with exclusions and prominent assumptions.
Elemental	Concept geometry, selected assemblies, approximate quantities, verified rates.	Work-item cost breakdown with rate dates and uncertainty.
Detailed BOQ	Verified drawings, measured quantities, specifications, professional review.	Tender-ready work items only after qualified measurement and approval.

Cost structure / codes

Codes	Cost groups
C01-C04	Site preparation; substructure; superstructure; external envelope
C05-C08	Internal construction; finishes; mechanical services; electrical
C09-C12	Plumbing; fire/specialist systems; external works; equipment
C13-C17	Labor/equipment not already in work rates; preliminaries; design/fees; contingency; taxes/escalation

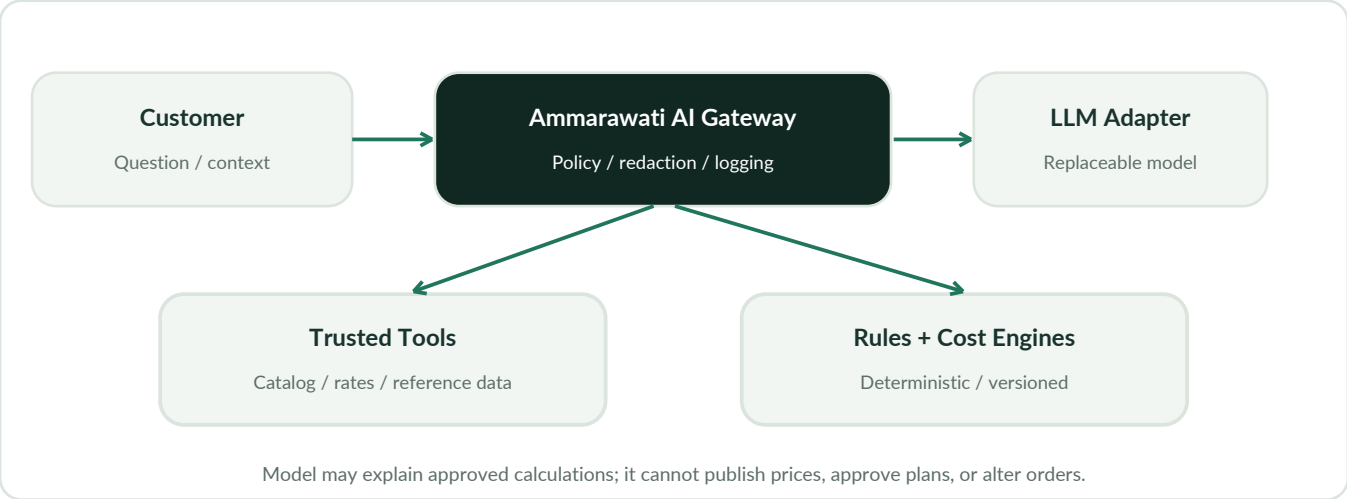
Calculation: Work item subtotal = measured quantity × approved unit rate. Project estimate = summed non-overlapping work-item costs + indirect allowances + professional/statutory costs + contingency + applicable tax/escalation. All currencies, unit conversions, inclusions/exclusions and rate sources must be explicit. Align cost measurement concepts with RICS NRM 1/2 where helpful, without assuming UK-specific local rates [R7].

Rate management workflow

Regional rate proposal -> validation -> reviewer approval -> effective rate publication -> audit/version snapshot. Every record includes material_id, region, quote/supplier source, unit, currency, amount, applicable dates, reviewer and state. Detect abnormal changes; publish only complete approved values. Existing estimates preserve historical prices.

Operator screen	Critical validation
Rate master	Specification, unit, currency, geography, evidence, duplicate and date checks.
Bulk import	Preview, reject invalid rows, dry run, change summary and rollback procedure.
Rate approval	Maker/checker separation; threshold-dependent approvals; immutable history.
Estimate report	Sources, last-rate dates, omissions, level/confidence, PDF export and review notice.

15 / Replaceable AI and future company intelligence



Model may explain approved calculations; it cannot publish prices, approve plans, or alter orders.

At launch, connect an approved hosted open-weight model API or a self-hosted model through an **AI Gateway**. The storefront never calls a model provider directly. The provider adapter can later be swapped without changing customer records or estimates.

Allowed model capabilities and restrictions

AI may help with	AI may NOT decide autonomously
Answer educational material questions using reviewed source documents	Publish/regenerate approved material prices
Clarify missing project inputs and summarize customer options	Approve structural designs or municipal compliance
Explain deterministic estimate inputs, alternatives and gaps	Commit purchases, refunds or stock changes
Translate English/Nepali technical content for review	Read unrelated customer records or admin secrets
Recommend informational next steps and professional review	Write unreviewed engineering rules to production

Future model migration strategy

- Use a single canonical request/response contract with provider-specific adapters.
- Keep permitted source documents, model version and prompt/policy version in auditable records.

- Run evaluation sets covering Nepal terminology, sourcing, refusals and numerical consistency.
- Restrict model tool calls to read-only allowlisted operations unless separately approved and audited.
- Minimize customer data sent to external providers; enforce procurement, privacy and retention controls.
- Apply NIST AI RMF guidance for testing and monitoring generative AI risk [R8].

16 / Admin UI, roles and review queues

Super Admin dashboard

Top KPI row: total sales, paid orders, active customers, product searches, new construction-project leads, requests awaiting approval. Filters: date range, region, channel and customer segment. Main body: sales funnel, popular search terms, price interest, no-result searches, regional demand, projects by type and security exceptions.

Regional Admin dashboard

Top KPI row: local paid orders, unfulfilled shipments, enquiries, pending quote reviews and proposed local rate changes. Main body: order queue, assigned leads, local category demand and alerts. No company-wide personal customer exports.

Action	Super Admin / approved specialist	Regional Admin
Global catalog / settings	Manage with scoped change controls	Read or propose only
Customer records	Authorized purpose-limited access	Assigned region only
Local rates	Publish / approve by privilege	Propose and justify
Orders	Company-wide ops by specific role	Assigned fulfillment region
Engineering references	Publish only after qualified verification	Read / flag issues
Security controls	Separate security role and MFA	No platform security changes
Sensitive export	Approval + logging + minimum scope	Usually aggregate only

Operational UX pattern: navigable left rail; global filters; searchable tables; status chips; entity-detail drawers; explicit confirmation for approvals; linked audit history; keyboard-friendly forms. All rights enforced by APIs, not only hidden buttons.

17 / Security, privacy and operational resilience

Control family	Baseline controls and tests
Identity	OIDC, strong MFA for staff, secure recovery, rate-limited auth, device/session revocation.
Access	RBAC plus region/ownership policies; negative tests for IDOR and cross-region access.
Application	OWASP ASVS 5.0.0-aligned controls; CSRF, XSS, injection prevention, safe headers [R5].
Data	Encrypted database/storage/backups; KMS/HSM managed keys; private network; secret rotation.
Payments	Hosted gateway, signed webhook verification, idempotency, no raw card details.
Delivery	SAST/DAST, dependency/SBOM, signed artifacts, reviewed migrations, scoped CI/CD.
Monitoring	Central audit trail, threat alerting, incident escalation, log retention and redaction.
Recovery	Automated encrypted backups, separate backup account/cloud option and restore drills.
Privacy	Consent/choice, purpose limitation, retention, access requests, analytics separation.
AI	Prompt-injection defenses, restricted tools, citations, redaction and evaluation tests.

SECURITY DESIGN POSITION

A “private” database means controlled infrastructure and narrow application access; it does not automatically provide safety. Define who holds keys, how changes are approved, how incidents are handled and how data can be restored.

18 / Cloud deployment and portability

Start with one company-controlled primary cloud organization with isolated dev/stage/prod environments, private network segments and a PostgreSQL primary plus appropriate high-availability/backup design. Use containers, reproducible infrastructure definitions and standardized secrets/object storage interfaces. A second cloud can later host encrypted backup copies, a disaster-recovery environment or selected stateless services.

Production zone	Components / controls
Public edge	DNS, CDN, WAF, DDoS protection, TLS termination and request limits.
Web application zone	Public storefront and both admin frontends; admin access policy strengthened.
Private application zone	API, workers, identity middleware, policy engine and AI Gateway.
Private data zone	PostgreSQL primary/standby as appropriate; restricted service connections.
Backup & recovery	PITR where supported, immutable/isolated backup copies, independent restore tests.
Observability zone	Logs/metrics/traces, alerting and tested incident runbooks.
Secondary cloud later	Tested backups and controlled cutover; no uncontrolled multi-writer commerce data.

Provisional quality targets: 99.9% service SLO, agreed workload P95 latency target, documented data-retention objectives, and a budgeted recovery design. Illustrative RPO = 15 minutes / RTO = 4 hours only if selected infrastructure and drills demonstrate it; no untested guarantees.

19 / Implementation roadmap and team handoff

For a secure commerce-plus-planning first release, use a staged baseline of roughly **28-36 weeks** with overlapping workstreams and an experienced team. A smaller catalog/lead-capture pilot can precede full checkout and planning. Scope, local data availability and integrations can materially change duration.

Phase / weeks	Main deliverables / exit criteria
0. Discovery (1-3)	Approved routes, business rules, site data, privacy decisions, threat model, prototype screens.
1. Foundations (3-7)	Repo, CI/CD, environments, identity, PostgreSQL migration, API contracts, design system.
2. Catalog & events (6-12)	Product import, search, category pages, published prices, guest cart, consent gate and event capture.
3. Commerce/admin (10-20)	Account checkout, payment, orders, stock; Super and Regional portals; RBAC negative tests.
4. Planning & rates (16-27)	Nepal plot input, template SVG concepts, rule registry, operator rates, 3-level estimating.
5. Knowledge & AI (20-30)	Reviewed material pages, AI Gateway, safety policies, response evaluation.
6. Audit / pilot (28-34)	Load, accessibility, penetration, reconciliation and disaster-restore testing.
7. Launch (34-36)	Support handover, monitoring, production checklist, on-call and documentation.

Recommended delivery team

Product owner/business analyst; technical architect; 2 backend engineers; 2 frontend engineers; UI/UX designer; DevOps/security engineer; QA automation tester; content manager; qualified Nepal architect/civil engineer for reviewed construction guidance. Assign explicit internal owners for legal, material rates, operations and customer support.

Definition of done

- Product search, visible prices, guest cart, checkout and server-confirmed payments function end to end.
- Regional administrators cannot access cross-region customer, order, lead or price records.
- Price changes require intended approval and preserve historical estimate versions.
- Consent controls prevent unauthorized optional tracking; deletion/access requests are testable.
- Unverified municipality rules result in an explicitly unverified concept, not fabricated setbacks.
- AI answers cannot modify approved prices or present sketches as safe construction drawings.
- Backups restore successfully, logs/alerts work, and critical/high security issues are closed.
- All sites are responsive, accessible and instrumented with operational health monitoring.

20 / Build backlog: first concrete tasks

ID / priority	Developer-ready task	Acceptance evidence
A01 / P0	Create monorepo, type checking, linting, CI quality gates	Three app shells + build pipeline
A02 / P0	Provision private PostgreSQL + migrations + backups	Private connection and restore verification
A03 / P0	Model products, variants, price versions and regions	Seed catalog and tested filters
A04 / P0	Build responsive category, search, product pages	Mobile/desktop UAT and accessibility tests
A05 / P0	Implement privacy notice, choice and event collector	Consent gating unit/integration tests
A06 / P0	Build identity, cart, checkout and payment state	Idempotent test-payment scenarios
A07 / P0	Build Super and Regional auth/permissions	Cross-region denial and audit proof
A08 / P1	Implement rate proposal, review and publication	Historical price version remains reproducible
A09 / P1	Implement land input and SVG concept templates	Verified geometry for supported rectangles
A10 / P1	Implement estimate engine and PDF summary	Deterministic reproducible result
A11 / P1	Add reviewed standards and educational content	Content approval/version checks
A12 / P1	Integrate model adapter through protected gateway	Swappable provider smoke tests

21 / Risk register, owner checklist and decisions

Risk / gap	Mitigation before activation
No verified material rates	Show “price unavailable”; operator import + reviewer publication.
Unknown municipality rules	Require rule-source validation or label outputs “unverified concept”.
Engineering over-reliance	Qualified technical reviewer; prominent limitations; no final designs.
Unauthorized tracking	Purpose-gated collection, regional legal review and visible user controls.
Checkout inconsistency	Transactional reservations, idempotent payment processing and reconciliation.
Cross-region data exposure	Permission matrix, API policies and automated negative authorization tests.
Vendor lock-in	Container portability, standard PostgreSQL, backups, provider adapters.
AI hallucinations	Retrieved verified facts, allowlisted tools, rule-backed calculations and evaluation.
Overbuilt first release	Keep phase-one commerce scope narrow and pilot land-planning templates.

Owner sign-off checklist

- Appoint product owner, cloud owner, security owner, pricing approver and qualified Nepal engineering reviewer.
- Approve two target municipality/region pilots and the first 100-500 real product SKUs (scope example, not a requirement).

- Choose official domain, cloud provider/region, payment provider and support model.
- Approve privacy wording, retention schedule and legal/customer-terms review.
- Prepare rate-data source contracts and verified technical material/content documents.
- Sign off on mobile homepage, product page, cart, two admin dashboards and basic land-planner prototypes.
- Start 2-week sprints with end-to-end demos and test evidence.

22 / Reference sources and verification notes

These sources inform the architecture and compliance references. They are not a substitute for local legal advice or qualified municipal/structural design review. Checked against publicly available pages on 26 October 2026. Links are clickable in the PDF.

[R1] The Builders Supply - reference e-commerce storefront
https://thebuilderssupply.com/
[R2] Nepal DUDBC - national building-code resources (NBC 205:2024; NBC 206:2024)
https://www.dudbc.gov.np/pages/24231978/
[R3] Kathmandu eBPS - government-hosted NBC 105:2025 PDF
https://ebps.kathmandu.gov.np/UploadFiles/NBC_105_2025_Sesmic_Design.pdf
[R4] Kathmandu Metropolitan City eBPS - building bylaws and amendments
https://ebps.kathmandu.gov.np/Hom/ReferenceDoc
[R5] OWASP Application Security Verification Standard 5.0.0
https://asvs.owasp.org/
[R6] Nepal Law Commission - Electronic Commerce Act 2081
https://lawcommission.gov.np/content/13517/electricity-trade--e-workers--act--2081/
[R7] RICS New Rules of Measurement (NRM 1, 2 and 3)
https://www.rics.org/profession-standards/rics-standards-and-guidance/sector-standards/construction-standards/nrm
[R8] NIST Generative AI Risk Management Profile (AI 600-1)
https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence

Document status: implementation guide only. It does not establish that the platform, database, cloud or engineered design has been deployed.